

Descriptive Statistics I

What Statistics Is, Data Types, Measurement Scales, Errors, Every Chart You Need & Central Tendency — with Python

CodeByPaxto: The AI & ML Playground

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What We're Covering Today

- 1 What statistics actually is, and the two types: descriptive vs. inferential
- 2 Population vs. sample, and dependent vs. independent variables
- 3 Collecting & measuring data — sources, sampling techniques, and reliability vs. validity
- 4 Types of data and the four measurement scales — and where each is used
- 5 Errors and bias in data — what's random, what's systematic, and how to reduce it
- 6 The complete chart & graph catalog — when to use each, and when not to
- 7 Measures of central tendency — mean, median, and mode
- 8 Today's tasks and a mini project — profiling a real dataset



Your First Python Cell

- You do NOT need to know these libraries yet — copy, run, and read the comments.
- math and statistics ship with Python.
numpy, pandas and scipy are one pip / conda install away.
- Every code slide today uses small, made-up data so you can run it anywhere.

```
• • •  
  
# Built into Python: nothing to install  
import math  
import statistics  
  
# One-time install: pip install numpy pandas scipy  
import numpy as np  
import pandas as pd  
from scipy import stats  
  
scores = [72, 85, 90, 66, 85]  
print(statistics.mean(scores)) # 79.6  
print(np.mean(scores))        # 79.6  
print(math.sqrt(16))           # 4.0
```


PART 1

What Is Statistics?

Before any formula, a clear picture of what this field actually does, and the two very different jobs it does it for



What Is Statistics, Really?



?

Making sense of data

- Statistics is the field of collecting, organizing, summarizing, and drawing conclusions from data
- At its core, it answers two very different kinds of questions: "what does this data show?" and "what can this data tell me about something bigger than itself?"
- Every dataset is meaningless until statistics gives you a way to describe it, compare it, or trust (or doubt) a conclusion drawn from it
- This is genuinely one of the most widely used branches of math in the world — not just in AI, but in every field that collects data of any kind



Where Statistics Is Actually Used

1

Medicine & Research

Clinical trials, drug effectiveness, and public health decisions all rest on statistical evidence

2

Business

Sales forecasting, customer segmentation, and A/B testing a new feature or price

3

Sports

Player performance metrics, game strategy, and predicting match outcomes

4

Government & Policy

Census data, unemployment rates, and evidence behind public policy decisions



Descriptive vs. Inferential Statistics

Descriptive Statistics

- Describes and summarizes the exact data you already have — nothing more
- Answers "what happened in THIS data?" — the average, the spread, the shape
- This entire session, and the next, live here

Inferential Statistics

- Uses a smaller sample to draw conclusions about a larger population you didn't fully measure
- Answers "what can I confidently CONCLUDE, beyond just this data?" — confidence, significance, hypothesis tests
- This is where the course goes in Sessions 21 and 22



Population vs. Sample



**The whole group, vs. a
slice of it**

- A population is every single member of the group you actually care about — every voter, every customer, every product ever made
- A sample is a smaller, manageable subset of that population, actually measured — usually because measuring the entire population is impossible or impractical
- Almost everything in real data science is a sample — you rarely have data on literally everyone or everything
- This single distinction is the entire reason inferential statistics needs to exist at all — keep it in mind, it comes back constantly



Sample vs. Population in Python

- We invent a population of 10,000 scores, then measure only 100 of them.
- The sample mean lands close to the population mean — but almost never exactly on it.
- That small gap is exactly why inferential statistics exists.

```
import numpy as np

rng = np.random.default_rng(42)
population = rng.normal(70, 10, size=10_000)

sample = rng.choice(population, size=100,
                    replace=False)

print(round(population.mean(), 2)) # 69.9
print(round(sample.mean(), 2))    # 71.11
print(len(population), len(sample)) # 10000 100
```



Dependent vs. Independent Variables



Cause, and the effect

- An independent variable is the one you change, control, or use to explain something — it stands on its own, unaffected by the other variables in the study
- A dependent variable is the one you measure as an outcome — its value "depends on" the independent variable
- In "does study time affect exam score," study time is independent, exam score is dependent — the score depends on the hours studied, not the other way around
- This distinction quietly decides which axis a variable belongs on in a scatter plot, and which side of a formula it belongs on in a regression — both coming up soon



Independent & Dependent Variables in a Table

- pandas stores data as a table: each column is one variable.
- Put the independent variable in x and the dependent variable in y — you will reuse this habit in every ML model.
- `df["hours"]` simply pulls out one column.

```
import pandas as pd

df = pd.DataFrame({
    "hours": [1, 2, 3, 4, 5],      # independent (x)
    "score": [52, 60, 58, 71, 75], # dependent (y)
})

x = df["hours"] # what we change or explain
y = df["score"] # what we measure as outcome

print(x.mean(), y.mean())      # 3.0 63.2
print(df.shape)                # (5, 2)
```



PART 1



Practice Set — What Is Statistics?

Try these yourself first — pencil, paper or a quick Python cell.

- 1 Name one real decision (personal, business, or public) that was likely made using statistics
- 2 For a survey of 200 university students about study habits, identify the population and the sample
- 3 Explain in your own words the difference between descriptive and inferential statistics, using your own example for each
- 4 For "does fertilizer amount affect plant height," identify the independent and dependent variable, and explain your reasoning

PART 2

Collecting & Measuring Data

Where your data actually comes from, how it was sampled, and whether the way it was measured can even be trusted



Primary vs. Secondary Data



Who collected it first

- Primary data is collected firsthand, specifically for your own purpose — your own survey, your own experiment, your own sensor readings
- Secondary data was collected by someone else, for a different original purpose, and you're reusing it — government census data, a public dataset, a previous study's results
- Primary data fits your exact question perfectly, but costs real time and money to collect. Secondary data is fast and often free, but may not perfectly match what you actually need
- Most real-world data science work is a mix of both — recognizing which kind you're holding shapes how much you should trust and reshape it



Sampling Types & Techniques



#

How you pick the slice

- How a sample is chosen directly determines whether it can honestly represent the population it came from
- Probability sampling gives every member of the population a known chance of being selected — this is what makes inferential statistics (Sessions 21–22) mathematically valid at all
- Non-probability sampling doesn't guarantee that — it's often faster and cheaper, but carries a real risk of a biased, unrepresentative sample
- The next slide walks through the specific techniques inside each category



Common Sampling Techniques

1

Simple Random

Every member has an exactly equal chance of selection — like drawing names from a hat

2

Stratified

The population is split into meaningful subgroups first, then randomly sampled within each

3

Cluster

The population is split into naturally occurring groups (e.g. schools), and whole groups are randomly selected

4

Convenience

Whoever is easiest to reach is sampled — fast, but carries the highest risk of an unrepresentative sample



More Sampling Techniques

1

Systematic

Select every k -th member from a list (e.g. every 10th name) after a random starting point — probability-based

2

Snowball

Existing participants refer other participants — useful for hard-to-reach populations, but non-random

3

Quota

A fixed number is sampled from each subgroup, like stratified — but selection within each group isn't random



Hybrid Sampling in Practice

%

**Mixing methods,
honestly**

- Real research often combines techniques rather than picking just one — for example, stratifying a national population by region, then using convenience sampling at accessible locations within each region
- This is completely legitimate — it balances practicality against representativeness far better than pure convenience sampling alone
- The key obligation is transparency: a hybrid approach should be reported exactly as it was done, so readers can judge for themselves how representative the sample really is
- Hiding a convenience step inside what looks like a fully random design is exactly how a study quietly inherits selection bias without anyone noticing



Three Sampling Techniques in Python

- Simple random: every row has an equal chance.
- Systematic: take every k-th row. Stratified: sample inside each group.
- Convenience and snowball sampling cannot be coded — they depend on how you reach people in the real world.

```
import pandas as pd

df = pd.DataFrame({
    "student": range(1, 101),
    "year": ["1st"] * 40 + ["2nd"] * 35 + ["3rd"] * 25,
})

simple = df.sample(n=10, random_state=1)
systematic = df.iloc[::10] # every 10th row
strat = df.groupby("year").sample(frac=0.1,
                                  random_state=1)

print(len(simple), len(systematic), len(strat))
# → 10 10 10
print(strat["year"].value_counts().to_dict())
# → {'1st': 4, '2nd': 4, '3rd': 2}
```



Probability vs. Non-Probability Sampling

Probability Sampling

- Simple random, stratified, cluster, and systematic sampling all belong here
- Every member has a known, nonzero chance of selection
- This is what lets you legitimately generalize a sample's result back to the full population

Non-Probability Sampling

- Convenience, snowball, and quota sampling all belong here
- Selection isn't random — some members may have no chance of being included at all
- A hybrid approach (mixing both) is common in practice — useful, but should be reported honestly, not hidden



Operationalization & Proxy Measurements



**Turning an idea into a
number**

- Operationalization is the process of turning a vague, abstract idea — "happiness," "customer loyalty," "intelligence" — into something concrete enough to actually measure
- A proxy measurement is a stand-in variable used because the real thing can't be measured directly — using "number of repeat purchases" to represent "customer loyalty"
- Every operationalization involves a judgment call, and a different one can lead to a genuinely different result — this is worth stating openly in any real analysis
- This is precisely the step where "reliability" and "validity," on the next slide, actually get put to the test



Reliability vs. Validity

Reliability

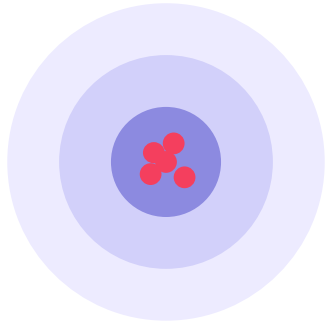
- Asks: if I measure this again, under the same conditions, do I get a consistent result?
- A bathroom scale that reads 3kg heavier every time is reliable — wrong, but consistently wrong
- Without reliability, you can't trust any single measurement at all

Validity

- Asks: am I actually measuring the concept I claim to be measuring?
- A survey meant to measure "job satisfaction" that actually measures "salary satisfaction" is NOT valid, even if it's perfectly reliable
- A measurement can be reliable without being valid — but it can't be genuinely valid without also being reliable

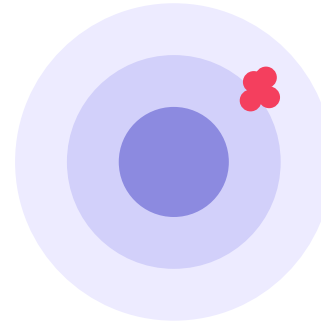


Reliability vs. Validity, Visualized



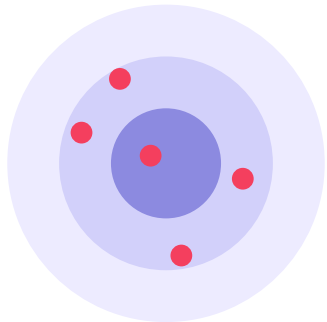
Reliable AND Valid

Tightly clustered, right on target



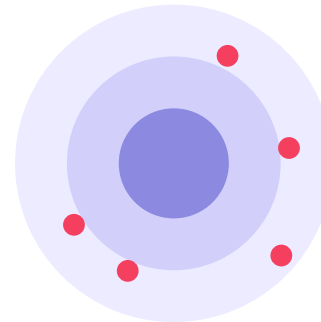
Reliable, but NOT Valid

Tightly clustered — consistently off-target



NOT Reliable, but Valid "on average"

Scattered widely, but centered on target



NOT Reliable, NOT Valid

Scattered randomly, nowhere near the target

Reliability asks: if I measure this again, do I get a consistent result? Validity asks: am I actually measuring the right thing? A measurement needs both to be trustworthy.



Reliability vs. Validity: Measure It

- Reliability = how small the spread is (standard deviation).
- Validity = how close the average is to the true value (bias).
- Each tuple below is (bias, spread) for 1,000 readings of a true value of 50.

```
import numpy as np

rng = np.random.default_rng(0)
true_value = 50

def check(m):
    bias = m.mean() - true_value # validity
    spread = m.std() # reliability
    return round(float(bias), 1) + 0, round(float(spread), 1)

good = rng.normal(50, 1, 1000) # reliable + valid
off = rng.normal(56, 1, 1000) # reliable, NOT valid
noisy = rng.normal(50, 8, 1000) # NOT reliable, valid

print(check(good)) # (0.0, 1.0)
print(check(off)) # (6.0, 1.0)
print(check(noisy)) # (-0.4, 7.9)
```



Practice Set — Collecting & Measuring Data

Try these yourself first — pencil, paper or a quick Python cell.

- 1 For your own final project idea, decide whether your data would be primary or secondary, and explain why
- 2 A researcher samples only students who walk past the library at noon. Name this sampling technique, and explain its main risk
- 3 A study on a rare medical condition asks each participant to refer other people with the same condition. Name this technique, and explain when it's genuinely the right choice despite being non-random
- 4 Propose one reasonable way to operationalize "student engagement" using a measurable proxy
- 5 A weight scale gives a different reading every time you step on it, seconds apart, with nothing else changing. Is this a reliability problem or a validity problem? Explain

PART 3

Types of Data & Measurement Scales

Not all data can be treated the same way — this determines which statistics and which charts are even valid to use



Categorical vs. Numerical Data

Categorical Data

- Describes a quality or category — groups, labels, names
- Nominal: no natural order (blood type, country, color)
- Ordinal: a natural order exists, but distances aren't meaningful (satisfaction rating, education level)

Numerical Data

- Describes a measurable quantity — counts or measurements
- Discrete: countable, whole values only (number of children, number of orders)
- Continuous: can take any value in a range (height, temperature, time)



Data Types in pandas

- pandas tells you each column's type with `.dtypes`.
- Ordinal data needs an explicit order — otherwise Python treats labels as plain text.
- An ordered Categorical lets you ask for max or min of a rating.

```
import pandas as pd

df = pd.DataFrame({
    "blood": ["A", "O", "B"],      # nominal
    "rating": ["low", "high", "mid"], # ordinal
    "kids": [0, 2, 1],            # discrete
    "height": [160.5, 172.0, 181.2], # continuous
})

df["rating"] = pd.Categorical(
    df["rating"], categories=["low", "mid", "high"],
    ordered=True)

print(df.dtypes)
print(df["rating"].max())      # high
```



The Four Measurement Scales, at a Glance

1

Nominal

Labels only — no order, no distance. Example: eye color, country of birth

2

Ordinal

Has order, but the gaps between values aren't equal. Example: movie star ratings

3

Interval

Equal gaps between values, but no true zero. Example: temperature in °C

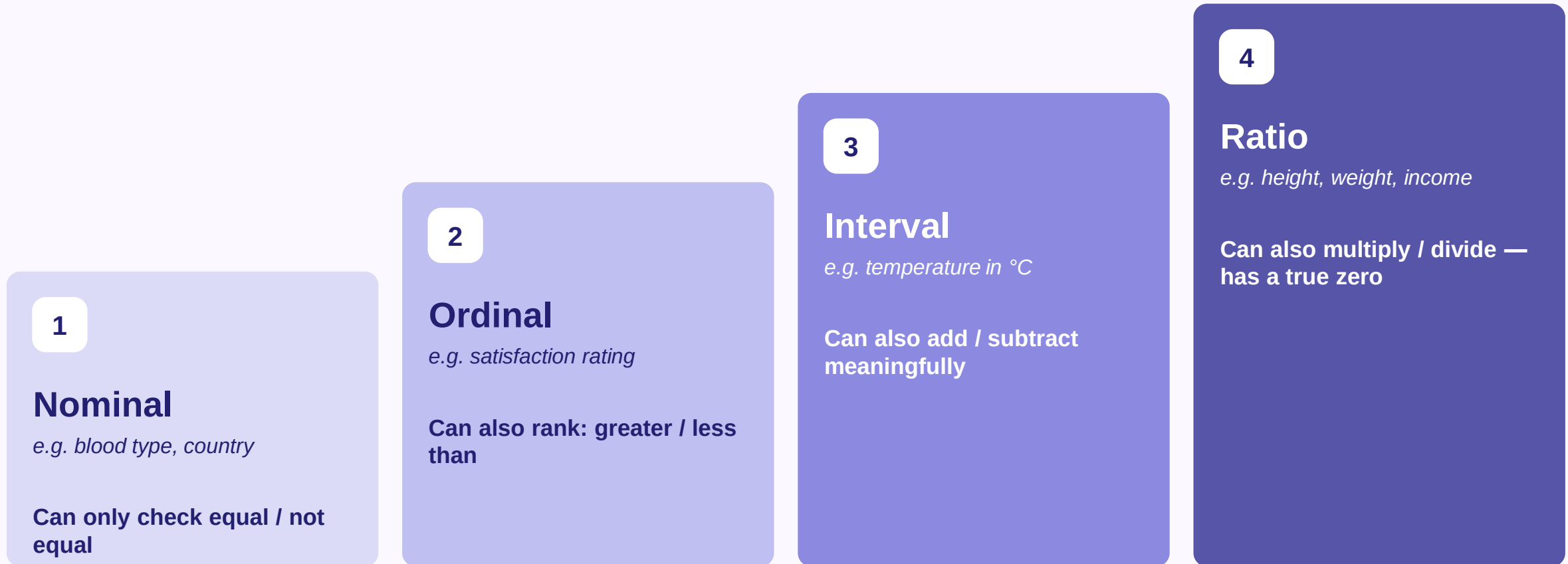
4

Ratio

Equal gaps AND a true zero, meaning "none at all." Example: weight, income, age



The Measurement Scales Ladder



Each step up the ladder keeps everything the step below could do, and adds one new ability on top of it.



Nominal Scale — Labels Only

μ

Just names, no order

- Nominal data sorts things into named categories, with no meaningful order between them — "red" isn't more or less than "blue"
- Examples: gender, blood type, country of residence, a customer's preferred payment method
- The only valid comparison is equality — you can say two things are the same category or different categories, and nothing more mathematically
- You can count how often each category appears, and find the mode — but calculating a "mean country" makes no sense at all



Ordinal Scale — Order, But No Real Distance



**Ranked, but not
measured**

- Ordinal data has a genuine, meaningful order — but the gap between adjacent values isn't guaranteed to be equal
- Examples: a 5-star rating, a t-shirt size (S/M/L/XL), an education level (high school/bachelor's/master's)
- The jump from 1 star to 2 stars might not represent the same "amount" of change as 4 stars to 5 stars — you only know the direction, not the exact size
- You can find the median and mode reliably — but averaging star ratings, while extremely common in practice, is technically stretching what the scale actually guarantees



Interval Scale — Equal Distance, No True Zero



**Zero doesn't mean
"none"**

- Interval data has consistent, equal gaps between values — the difference between 10°C and 20°C is the same "amount" as between 20°C and 30°C
- But zero doesn't mean "nothing" — 0°C isn't "no temperature at all," it's just the freezing point of water, an arbitrary reference
- Because of this, ratios are meaningless here: 20°C is NOT "twice as hot" as 10°C — that claim doesn't survive converting to Fahrenheit
- Classic examples: temperature in Celsius or Fahrenheit, and calendar year (year 0 doesn't mean "no time has passed")



Why Ratios Break on Interval Data

- 20°C to 40°C looks like “twice as hot” — it is not.
- Convert the same temperatures to Fahrenheit and the ratio changes.
- A true zero is what makes a ratio meaningful.

```
celsius = [20, 40]
fahrenheit = [c * 9 / 5 + 32 for c in celsius]

print(fahrenheit)           # [68.0, 104.0]
print(celsius[1] / celsius[0]) # 2.0
ratio_f = fahrenheit[1] / fahrenheit[0]
print(round(ratio_f, 2))     # 1.53

# Ratio data (money) survives any unit change
usd = [20, 40]
print(usd[1] / usd[0])      # 2.0
```



Ratio Scale — Equal Distance AND a True Zero



The most flexible scale

- Ratio data has everything interval data has, PLUS a true, meaningful zero — zero really does mean "none at all"
- Because of that true zero, ratios genuinely make sense: \$20 really is twice as much money as \$10; 40kg really is twice as heavy as 20kg
- Examples: height, weight, age, income, distance, time elapsed — most of the "hard numbers" you'll work with in data science live here
- Every mathematical operation is valid on this scale — which is exactly why it sits at the top of the ladder



Which Operations Are Valid on Which Scale

1

Nominal

Valid: mode, frequency counts. NOT valid: mean, median, or any ranking

2

Ordinal

Valid: mode, median, ranking. Debatable: mean (common in practice, technically imprecise)

3

Interval

Valid: mean, median, mode, addition/subtraction. NOT valid: meaningful ratios ("twice as much")

4

Ratio

Valid: absolutely everything — mean, median, mode, ratios, and every arithmetic operation



Valid Statistics for Each Scale

- Match the statistic to the scale before you compute anything.
- `statistics.mode` works on labels; median needs an order; mean needs real distances.
- Computing a “mean colour” is possible in code — but meaningless.

```
import statistics

colors = ["red", "blue", "red", "green"] # nominal
sizes = [1, 2, 2, 3, 4]                 # ordinal
temps = [20, 25, 30]                   # interval
income = [20, 40]                       # ratio

print(statistics.mode(colors)) # red
print(statistics.median(sizes)) # 2
print(statistics.mean(temps))  # 25
print(income[1] / income[0])   # 2.0
```



Practice Set — Types of Data & Measurement Scales

Try these yourself first — pencil, paper or a quick Python cell.

- 1 Classify each as nominal, ordinal, interval, or ratio: shoe size, exam grade (A/B/C), income, year of birth
- 2 Explain why "average zip code" is a meaningless calculation, using what you learned about nominal data
- 3 Explain why saying "40°C is twice as hot as 20°C" is incorrect, using the interval scale's lack of a true zero
- 4 For a customer satisfaction survey using a 1–5 star rating, identify the scale and explain what operations are safely valid on it

PART 4

Errors & Bias in Data & Measurement

Before trusting any statistic, it's worth asking where the data could have gone wrong on the way to you



Why Errors Matter Before You Even Start Analyzing



Garbage in, garbage out

- No formula can fix data that was collected badly — statistics can describe your data perfectly and still describe the wrong thing entirely
- Understanding where error creeps in is exactly what separates a careful analysis from a misleading one, especially in research
- Error, here, doesn't mean "a mistake someone made" — it means any gap between what you measured and the real truth, even an entirely honest one
- Two broad categories cover almost every real-world case: sampling error, and non-sampling error



Sampling Error

?

The sample wasn't the population

- Sampling error is the natural, unavoidable difference between a sample's result and the true population value — simply because you didn't measure everyone
- This exists even with a perfectly random, perfectly executed sample — it's a mathematical consequence of sampling itself, not a mistake
- Generally, larger samples have smaller sampling error — this is exactly why survey sample sizes matter so much
- This concept is the direct seed of "margin of error" and confidence intervals, which get fully covered in Session 21



Sampling Error Shrinks with Bigger Samples

- For each sample size we draw 200 samples and average how far off they land.
- Bigger samples sit closer to the true mean.
- numpy's `rng.choice` draws the random samples.

```
import numpy as np

rng = np.random.default_rng(1)
population = rng.normal(70, 10, 10_000)
true_mean = population.mean()

def avg_error(n):
    errs = []
    for _ in range(200):
        s = rng.choice(population, n)
        errs.append(abs(s.mean() - true_mean))
    return round(float(np.mean(errs)), 2)

print(avg_error(10))      # 2.78
print(avg_error(100))     # 0.8
print(avg_error(1000))    # 0.23
```



Non-Sampling Error — Everything Else That Can Go Wrong



**Avoidable, if you're
careful**

- Unlike sampling error, non-sampling error is NOT unavoidable — it comes from how the data was collected, entered, or reported, and better process can reduce it
- Measurement error: a faulty scale, a poorly worded survey question, or an inconsistent way of recording something
- Non-response error: the people who didn't answer your survey may differ systematically from those who did
- Data entry error: simple typos, duplicate records, or a unit mismatch (recording kg as lbs) during data collection



Common Sources of Non-Sampling Error, at a Glance

1

Measurement Error

The tool, question, or process used to record the data was flawed or inconsistent

2

Non-Response Error

The people who didn't respond likely differ from those who did, skewing results

3

Data Entry Error

Typos, duplicates, and unit mismatches introduced while recording or storing data



Catching Data Entry Errors with pandas

- Duplicates and impossible values are the most common data entry errors.
- `.duplicated()` flags repeated rows; a simple filter flags impossible ones.
- Here 1.82 looks like metres typed into a centimetres column.

```
import pandas as pd

df = pd.DataFrame({
    "name": ["Ali", "Sara", "Sara", "Omar", "Zoya"],
    "height": [170, 165, 165, 1.82, 999], # cm
})

print(df.duplicated().sum()) # 1

bad = df[(df["height"] < 100) | (df["height"] > 250)]
print(bad["name"].tolist()) # ['Omar', 'Zoya']

clean = df.drop_duplicates()
print(len(clean)) # 4
```



Bias — Not the Same Thing as Error



Systematic, not random

- Random error scatters your data around the true value — sometimes too high, sometimes too low, roughly canceling out on average
- Bias pushes your data consistently in ONE direction — it doesn't cancel out no matter how much data you collect, because the flaw is systematic, not random
- This is exactly why sampling error shrinks with a bigger sample, but bias does not — a biased sampling method just gives you a more precisely wrong answer with more data
- Recognizing bias means looking critically at HOW data was collected, not just how much of it there is



Common Types of Bias

1

Selection Bias

The sample systematically favors certain members of the population over others

2

Confirmation Bias

Collecting or interpreting data in a way that favors what you already expected to find

3

Response Bias

People answer inaccurately, often to appear more socially acceptable than their true answer

4

Survivorship Bias

Only the "survivors" of a process get measured, hiding the failures that dropped out



Bias Does Not Shrink with More Data

- A random sample of 200 lands near the true mean.
- A biased sample — only top performers — misses it by a lot, and collecting more would not help.
- Fixing bias means fixing the sampling design, not the sample size.

```
import numpy as np

rng = np.random.default_rng(3)
pop = rng.normal(70, 10, 10_000)

random_sample = rng.choice(pop, 200)
biased_sample = pop[pop > 75][:200] # top scorers only

print(round(pop.mean(), 1))      # 70.0
print(round(random_sample.mean(), 1)) # 69.2
print(round(biased_sample.mean(), 1)) # 81.0
```



How to Actually Reduce Bias



**Design it out, before
collecting**

- Randomization: using genuine probability sampling gives every member a fair, known chance of selection, directly countering selection bias
- Standardized instruments: using the exact same questions, wording, and measurement process for everyone reduces measurement and response bias
- Blinding: keeping data collectors unaware of the hypothesis being tested reduces the chance their expectations influence the results
- Pilot testing: trying your data collection process on a small group first surfaces confusing questions or flawed measurements before they contaminate the real dataset



PART 4



Practice Set — Errors & Bias

Try these yourself first — pencil, paper or a quick Python cell.

- 1 A survey about phone usage was only sent by email — explain what kind of error or bias this risks, and why
- 2 Explain, in your own words, why bias doesn't shrink with a larger sample size, while sampling error does
- 3 A fitness study only reports results from participants who completed the full 12-week program. Name this type of bias, and explain what's being hidden
- 4 Suggest one practical way to reduce response bias in a survey about employee satisfaction with their manager

PART 5

The Complete Chart & Graph Catalog

A picture is only worth a thousand words if it's the right picture — here's how to choose one on purpose



Why Chart Choice Actually Matters



#

**The wrong chart
misleads**

- Every chart makes a silent claim about your data's type and structure — using the wrong one doesn't just look odd, it can genuinely mislead your reader
- The right question is never "which chart looks nice" — it's "what is this data's type, and what relationship am I trying to show?"
- This catalog is organized around exactly that question: what your data looks like, first, then which chart fits it
- Every chart on the next several slides will be classified by three things: what it shows, when to use it, and when NOT to



Bar Chart: What It's For

Bar Chart

Data: categories + one number

WHAT IT'S FOR

Compares a number (a count, total or average) across separate categories, so you can instantly see which groups are bigger or smaller.

✓ Use it when

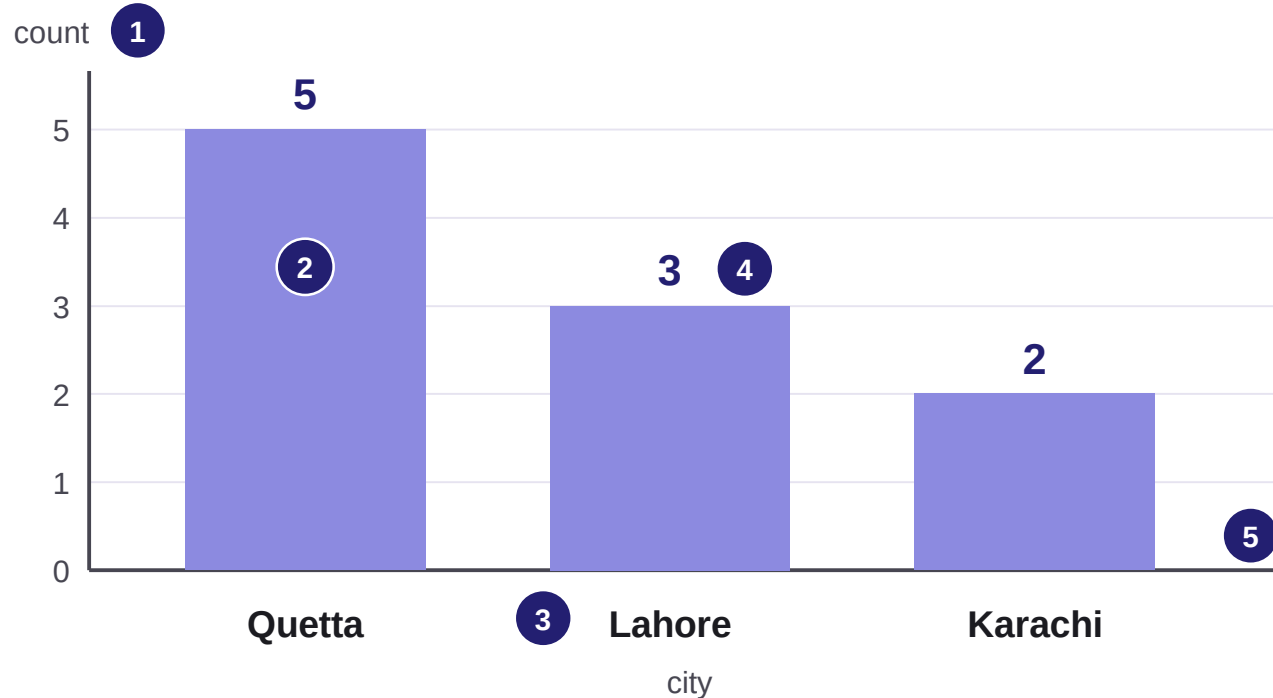
- Comparing counts or values across groups, such as students per city or sales per product
- Your x-axis is a category (nominal or ordinal), not a continuous number line
- You want to rank categories: sort the bars from tallest to shortest to make it obvious

✗ Avoid it when

- Showing change over time: a line chart shows the trend and direction far better
- Your variable is a continuous number like age or height: use a histogram to see its shape
- There are dozens of categories: the bars get cramped and the labels unreadable



Bar Chart: Anatomy & How to Read It



Components

- 1 Y-axis (value scale)**
What is measured: counts, totals or averages.
- 2 Bars**
One per category. The height shows its value.
- 3 X-axis (categories)**
Names each group. Gaps mean categories are separate.
- 4 Value labels**
The exact number on each bar, so nobody has to guess.
- 5 Zero baseline**
Bars must start at 0, or differences get exaggerated.

How to read it: Check the x-axis to see what each bar is, then read its height against the y-axis. Here Quetta has 5, Lahore 3 and Karachi 2, so Quetta is the most common. Compare bar heights, and make sure the axis starts at 0.



Histogram: What It's For

Histogram

Data: one continuous variable

WHAT IT'S FOR

Shows how the values of one continuous variable are distributed, by sorting them into equal-width ranges (bins) and counting how many land in each.

✓ Use it when

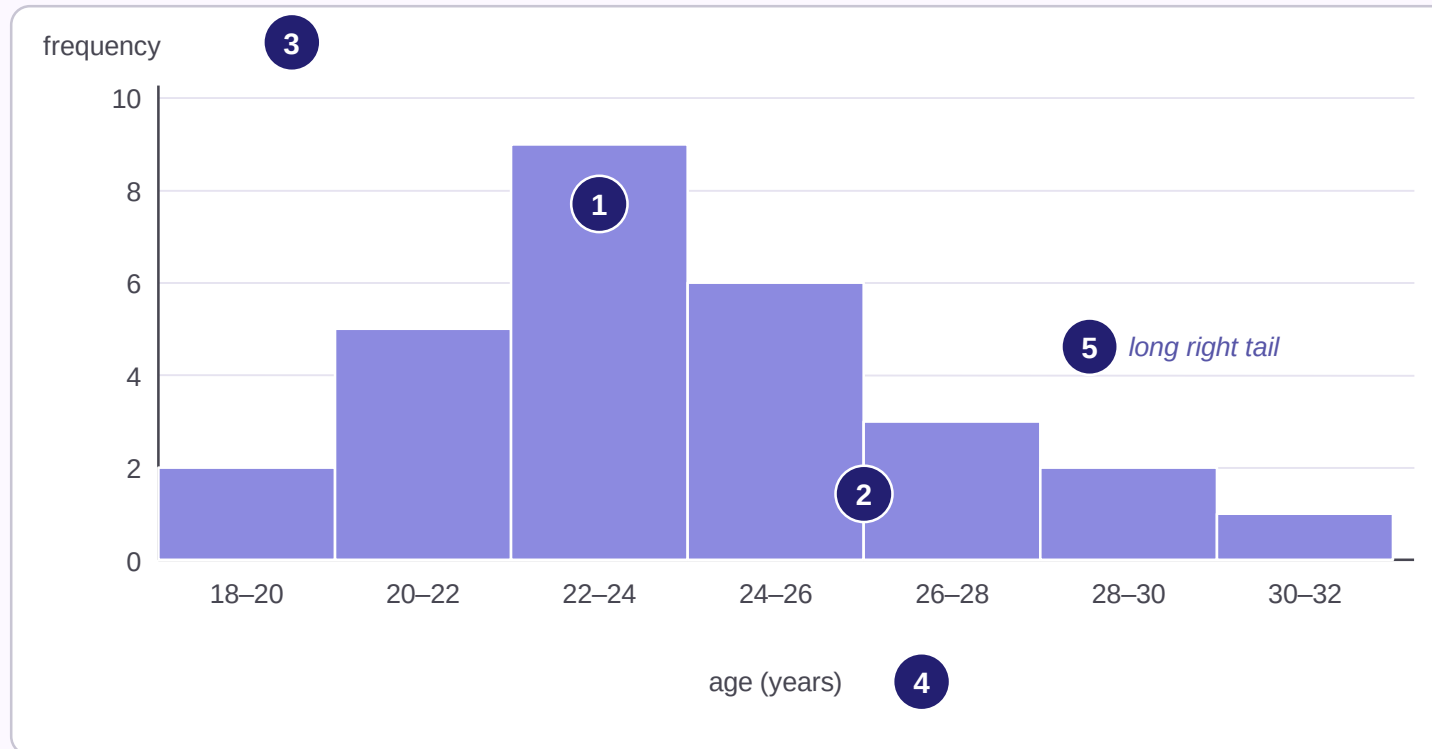
- Seeing the shape of a numerical variable: symmetric, skewed, or with more than one peak
- You have a decent amount of continuous data (roughly 30+ values), such as ages or waiting times
- Checking for gaps, clusters or odd values before choosing between the mean and the median

✗ Avoid it when

- The data is categorical: use a bar chart, because touching bars imply a continuous scale
- You only have a handful of values: the shape is just noise, so list the values instead
- You need to compare many groups: overlapping histograms get messy, so use box plots



Histogram: Anatomy & How to Read It



Components

- 1 Bars (bins)**
Each bar is a range of values; its height is how many fall inside.
- 2 No gaps**
Bars touch because the scale is continuous, unlike a bar chart.
- 3 Y-axis (frequency)**
The number of values in each bin. It starts at 0.
- 4 X-axis (value ranges)**
The variable, cut into equal-width bins. Here, age in years.
- 5 Overall shape**
The outline shows the peak, the spread and any skew.

How to read it: Find the tallest bar to see the most common range, here ages 22–24. Then check whether the outline is symmetric or has a long tail on one side. Gaps or lonely bars far from the rest may be outliers. Bin width changes the picture, so try a few.



Box Plot: What It's For

Box Plot

Data: one numerical variable

WHAT IT'S FOR

Boils one numerical variable down to five numbers (min, Q1, median, Q3, max) and flags outliers, so spread and centre can be compared at a glance.

✓ Use it when

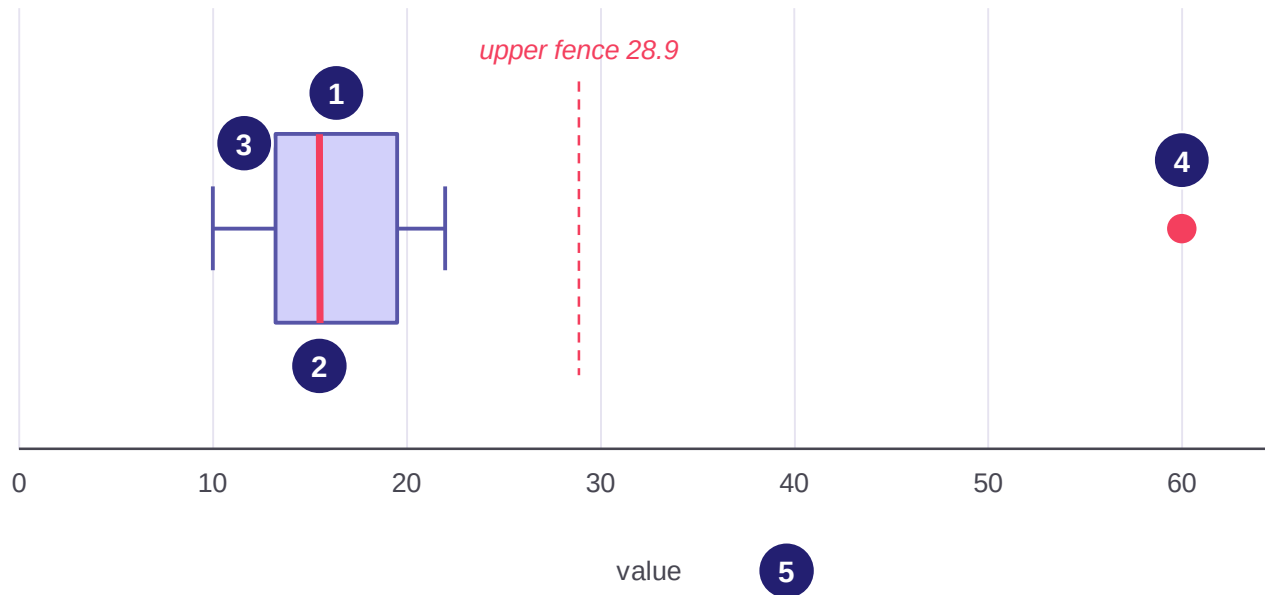
- Comparing the spread and centre of a numerical variable across several groups side by side
- Spotting outliers quickly, since they are drawn as separate dots beyond the whiskers
- You want a compact summary (median, quartiles, range) without plotting every value

✗ Avoid it when

- The exact shape matters: a box plot hides multiple peaks, so use a histogram
- You have very few values (under about 8): quartiles are unreliable, so show every point
- Your audience has never seen quartiles: use a simpler chart or explain it first



Box Plot: Anatomy & How to Read It



Data: 12, 15, 14, 10, 18, 20, 22, 16, 13, 60

Components

- 1 Box**
Runs from Q1 to Q3: the middle 50% of values (the IQR).
- 2 Median line**
The middle value: half the data lies below it, half above.
- 3 Whiskers**
Reach the smallest and largest values within $1.5 \times \text{IQR}$ of the box.
- 4 Outlier dot**
A value beyond the fence is drawn alone. Here, 60.
- 5 Axis**
The numeric scale, in the variable's own units.

How to read it: Start at the median line, then look at the box. A median near the middle of the box means a fairly symmetric middle half; near one edge means skew. Here the median is 15.5, the box covers 13.25 to 19.5, and 60 sits far beyond the upper fence of 28.9, so it is flagged as an outlier.



Scatter Plot: What It's For

Scatter Plot

Data: two numerical variables

WHAT IT'S FOR

Plots every observation as a dot using two numerical variables, revealing the direction, strength and shape of the relationship between them.

✓ Use it when

- Checking whether two numerical variables move together, like study hours and exam score
- Judging direction (positive or negative), strength, and whether the pattern is straight or curved
- Spotting clusters and outliers, or as a first look before computing a correlation

✗ Avoid it when

- One of the variables is a category: use a bar chart or a box plot by group instead
- You have huge datasets: thousands of dots pile into a blob, so sample or use a heatmap
- You are tracking one measure over time: a line chart is the right tool



Scatter Plot: Anatomy & How to Read It



Components

- 1 Y-axis**
The outcome you measure (dependent): exam score.
- 2 X-axis**
What you change or explain (independent): study hours.
- 3 Points**
One observation each: one student's hours and score.
- 4 Pattern**
Rising = positive; falling = negative; no shape = none.
- 5 Trend line**
Dashed line summarising the pattern. Dots hugging it = strong.

How to read it: Pick any dot, read across for its x value and up for its y value. Then look at the whole cloud: it rises from left to right and hugs the line, so more study hours go with higher scores, a fairly strong positive relationship. A relationship alone does not prove that one causes the other.



Heatmap: What It's For

Heatmap

Data: a grid of numbers

WHAT IT'S FOR

Turns a grid of numbers into colours, so patterns jump out. Most often used as a correlation matrix, showing how strongly every pair of variables is related.

✓ Use it when

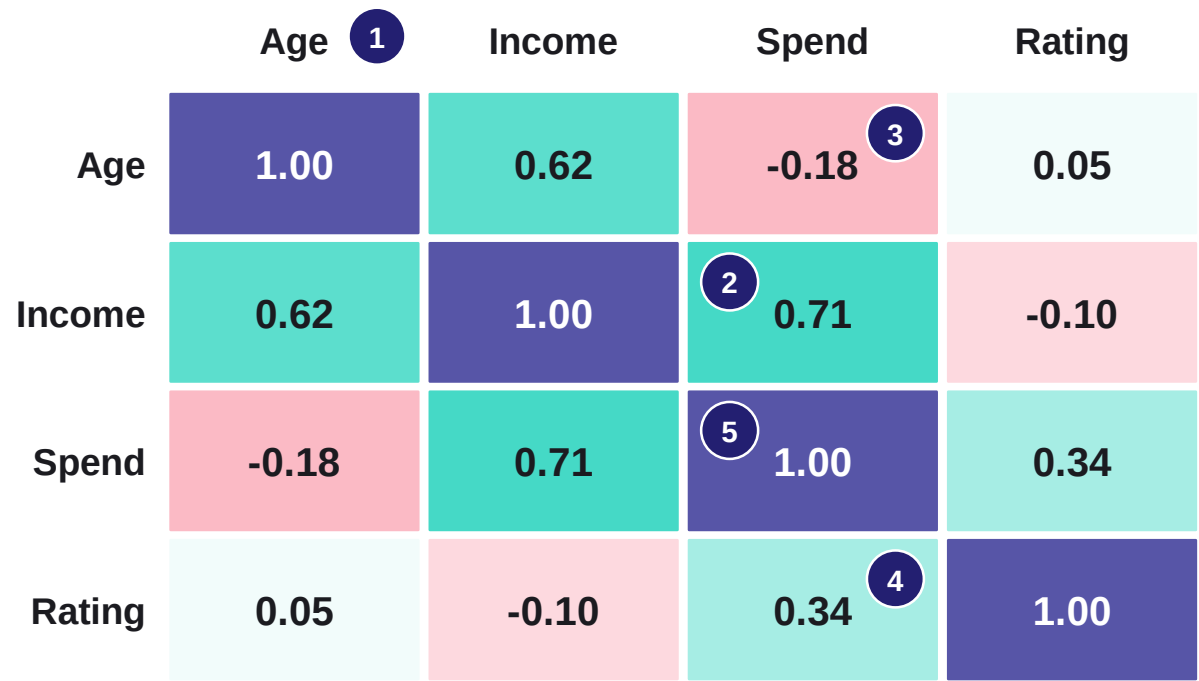
- Scanning relationships between many numerical variables at once, as a correlation matrix
- Any full grid of values, such as counts by weekday and hour, or a confusion matrix
- Finding which pairs deserve a closer look with a scatter plot

✗ Avoid it when

- You only have two or three variables: a scatter plot is far more precise
- Exact values matter: colour is hard to read precisely, so print the numbers or use a table
- The colour scale is confusing (rainbow palettes, no legend): readers will misjudge the values



Heatmap: Anatomy & How to Read It



Components

- 1 Rows & columns**
Every variable appears on both sides, so each pair meets in one cell.
- 2 Cells**
Each cell holds one pair's value, e.g. Income–Spend = 0.71.
- 3 Colour**
Teal = positive, pink = negative, pale = near 0.
- 4 Printed numbers**
The exact value in each cell, from -1 to +1.
- 5 Diagonal**
A variable with itself is always 1.00. Grid mirrors across it.

How to read it: Find a row, then a column: the cell where they meet is that pair's correlation. Look for cells far from zero. Income and Spend (0.71) rise together, while Age and Spend (-0.18) have only a weak negative link. Read just one triangle, since the grid is mirrored.



Common Chart Mistakes to Avoid

Misleading by Design

- A pie chart with 12+ tiny slices — nobody can compare them accurately
- A bar chart whose y-axis doesn't start at zero — exaggerates small differences dramatically
- A 3D chart adding visual flair but no real information — actively harder to read accurately

Wrong Chart for the Data

- A line plot connecting unrelated categories — implies a trend that doesn't exist
- A regression line forced through an obviously curved relationship
- Choosing a chart because it "looks impressive," rather than because it's genuinely the clearest way to show this specific data



PART 5



Practice Set — Choosing the Right Chart

Try these yourself first — pencil, paper or a quick Python cell.

- 1 For monthly sales over the last two years, name the best chart and explain why
- 2 For comparing the market share of 3 competing products, choose between a pie chart and a bar chart, and justify it
- 3 For checking whether study hours and exam scores are related, name the chart you'd reach for first
- 4 Explain what specifically is misleading about a bar chart whose y-axis starts at 80 instead of 0

PART 6

Measures of Central Tendency

The single number that tries to represent an entire dataset — and the three different ways to pick one



What Is Central Tendency?



**One number, standing in
for all**

- Central tendency is the general idea of finding a single, typical value that represents an entire dataset
- There isn't one universally "correct" way to do this — mean, median, and mode each answer a slightly different question
- Choosing the right one isn't a technicality — the wrong choice can genuinely misrepresent what your data actually looks like
- This choice depends directly on the measurement scale and shape of your data — exactly what the last two parts of this session set up



Mean, Median & Mode from Scratch

- Pure Python — no libraries — so you see exactly what each one does.
- Mean: sum divided by count. Median: middle of the sorted list. Mode: most frequent value.
- For an even-length list the median is the average of the two middle values.

```
data = [4, 8, 6, 5, 3, 8, 9]

mean = sum(data) / len(data)

s = sorted(data)
mid = len(s) // 2
median = (s[mid] if len(s) % 2
          else (s[mid-1] + s[mid]) / 2)

mode = max(set(data), key=data.count)

print(round(mean, 2))      # 6.14
print(median)              # 6
print(mode)                # 8
```



The Mean — The Arithmetic Average

- The mean adds up every value and divides by how many values there are — the formula from last session's notation lesson, in action
- It uses every single value in the calculation, which makes it sensitive — one extreme value can shift it substantially
- Only meaningful on interval or ratio data — calculating a "mean" of nominal categories makes no sense at all

$$\bar{x} = (1/n) \cdot \sum x_i$$

Example: [4, 8, 6, 10, 2]

$$\begin{aligned}\text{mean} &= (4 + 8 + 6 + 10 + 2) / 5 \\ &= 30 / 5 \\ &= 6\end{aligned}$$



The Median — The True Middle Value

- The median is the exact middle value once all your data is sorted — half the values fall below it, half above
- For an even number of values, it's the average of the two middle numbers
- It ignores how extreme the highest and lowest values are — this is exactly why it resists outliers so much better than the mean



Odd count: sort, take the exact middle

`[2, 4, 6, 8, 10] → median = 6`

Even count: average the two middle values

`[2, 4, 6, 8] → median = (4 + 6) / 2 = 5`

Works even on ordinal data (e.g. star ratings)



The Mode — The Most Frequent Value

- The mode is simply whichever value appears most often in the dataset
- It's the only measure of central tendency that works on nominal (category) data — mean and median don't apply there at all
- A dataset can have one mode, several (multimodal), or none at all if every value appears equally often



```
[2, 4, 4, 6, 8] → mode = 4
```

```
# appears twice
```

```
['red', 'blue', 'red', 'red']
```

```
→ mode = 'red'
```

```
[1, 2, 3, 4] → no mode
```

```
# every value appears once
```



The Same Three, with Libraries

- Every library gives the same answer — pick whichever is already in your code.
- statistics is simplest, numpy and pandas are fastest on big data.
- scipy.stats.mode returns the mode too, with its count.

```
import statistics
import numpy as np
import pandas as pd
from scipy import stats

data = [4, 8, 6, 5, 3, 8, 9]

print(statistics.mean(data))    # 6.142857142857143
print(np.median(data))         # 6.0
print(pd.Series(data).mode()[0]) # 8
print(stats.mode(data).mode)   # 8
print(statistics.multimode([1, 1, 2, 2, 3])) # [1, 2]
```



When the Mean Lies to You

The Classic Example

- Salaries at a small company: \$40k, \$42k, \$45k, \$48k, \$500k (the owner)
- Mean = \$135k — technically correct, but describes almost NOBODY'S actual salary
- Median = \$45k — this is what a "typical" employee genuinely earns

The Rule of Thumb

- When data is skewed or has extreme outliers, prefer the median
- When data is roughly symmetric with no extreme values, the mean is fine, and uses more information
- This is exactly why income, house prices, and similar data are almost always reported as medians in the news



When the Mean Lies — in Code

- One \$500k salary drags the mean to \$135k; the median stays at \$45k.
- A trimmed mean drops a slice from each end before averaging.
- A weighted mean counts some values more than others.

```
import numpy as np
from scipy import stats

salaries = [40, 42, 45, 48, 500] # in $k

print(np.mean(salaries))      # 135.0
print(np.median(salaries))    # 45.0
print(stats.trim_mean(salaries, 0.2)) # 45.0

# Weighted mean: exam worth 3x the quiz
print(np.average([80, 90], weights=[1, 3])) # 87.5
```



PART 6



Practice Set — Central Tendency

Try these yourself first — pencil, paper or a quick Python cell.

- 1 Calculate the mean, median, and mode of [5, 7, 7, 9, 100] by hand
- 2 Explain which measure best represents that dataset's "typical" value, and why
- 3 For a dataset of shoe sizes sold in a store, explain which measure of central tendency is most useful for restocking decisions
- 4 Find one real, published statistic that uses the median instead of the mean, and explain why that was likely the right choice

INDUSTRY-STYLE TASKS

Build the Judgment Every Real Analysis Depends On

Not abstract classification drills — choosing the right scale, chart, and average is what separates a clear analysis from a misleading one



TASKS



Part A: Statistics Basics & Variables

Small tasks — do them in order.

1

Write a two-sentence explanation, for a non-technical friend, of the difference between descriptive and inferential statistics, using your own example for each.

2

For a study on "does sleep duration affect reaction time," identify the population, a reasonable sample, and the independent and dependent variables.

3

Explain why almost all real-world data science analysis works with a sample rather than a full population, using a concrete example.



TASKS



Part B: Data Collection & Measurement Quality

Small tasks — do them in order.

1

For your own final project idea, decide whether you'd use primary or secondary data, and design a probability-based sampling plan to collect it.

2

Propose an operationalization with a measurable proxy for an abstract concept of your choice (e.g. "brand trust," "team morale").

3

Describe a realistic measurement tool or process, and evaluate it separately for reliability and for validity, explaining your reasoning for each.



TASKS



Part C: Data Types & Measurement Scales

Small tasks — do them in order.

1

Take any real dataset (or a provided one) and classify every single column by its measurement scale — nominal, ordinal, interval, or ratio.

2

For each ordinal column found, explain specifically why averaging it directly would be mathematically imprecise.

3

Write a one-paragraph explanation, for a non-technical reader, of why a "true zero" matters so much for ratio data.



TASKS



Part D: Errors, Bias & Chart Selection

Small tasks — do them in order.

1

Describe a realistic data collection scenario for your own final project idea, and identify one likely source of random error and one of systematic bias.

2

Given a dataset with 3 numerical columns, choose and justify one appropriate chart for each column individually, and one for showing their relationships together.

3

Find one real published chart (news article or report) that violates a "when NOT to use" rule from today, and explain specifically what's misleading about it.



TASKS



Part E: Central Tendency

Small tasks — do them in order.

- 1 Calculate the mean, median, and mode of a real numerical column from a dataset of your choice, by hand or with simple code.
- 2 Identify whether that column is skewed by extreme values, and explain which measure most honestly represents it.
- 3 Write a two-sentence summary of that column's "typical value," suitable for someone who has never seen the raw data.

MINI PROJECT

Put It All Together

Profile a real dataset from the ground up, using every skill from this session



Mini Project: Profile a Real Dataset



**Every skill from today,
combined**

- Choose any small, real dataset (or a provided one), and classify every column by data type and measurement scale
- State whether the data is primary or secondary, and identify a plausible source of both random error and systematic bias in how it might have been collected
- For three different columns, choose and justify an appropriate chart — one categorical, one numerical distribution, and one relationship between two variables
- Calculate mean, median, and mode for the key numerical column, and write a short, honest summary of what "typical" looks like in this dataset



Mini Project Starter: Profile a Dataset

- Start from this tiny table, then swap in your own dataset.
- Check the shape, the types, and the missing values before any statistic.
- `describe()` gives a first summary of every numeric column.

```
import pandas as pd

df = pd.DataFrame({
    "city": ["Quetta", "Lahore", "Quetta", "Karachi"],
    "rating": ["low", "high", "mid", "high"],
    "age": [21, 25, 22, None],
    "income": [30, 52, 41, 38],
})

print(df.shape)           # (4, 4)
print(df["age"].isna().sum()) # 1
print(df["income"].mean())  # 40.25
print(df["city"].mode()[0]) # Quetta
print(df.describe())
```

RECAP

Know Your Data First

Before any statistic means anything, you need to know your data's type, where it could have gone wrong, and the right picture to show it with. Next up: Descriptive Statistics II — spread, shape, z-scores and correlation.